V.F.S. v2.0 (Open-Gate)
Lyapunov Core Proof Formulae

with full term-by-term derivation and Sophianic Folding compatibility

Core extraction from `lyapunov.html`

Abstract

This hybrid document collects the core formulae of the Lyapunov proof for V.F.S. v2.0 (Open-Gate), preserves the original proof structure, and adds Sophianic Folding as a conservative Lyapunov-compatible refinement. It contains no graphs. It records the base dynamics, normalized variables, Lyapunov functional, active-domain walls, radial corridor, bounded Open-Gate source, memory bounds, non-Zeno estimate, and asymptotic cleansing statements. In addition, the central dissipative estimate $\dot{\mathcal{L}}_{\mathrm{VFS}} \leq C - C_1\mathcal{L}_{\mathrm{VFS}}$ — previously stated as a boxed result — is here *derived term by term*: the contraction rate C_1 is built from exactly two sign-definite blocks (resistance cleansing and will–action alignment), while the Sophia, radial and Open-Gate contributions are bounded and enter only the constant C. The auxiliary light-filter constant is proved positive in closed form, so that $C_1 > 0$ holds for *every* admissible parameter set under the standing walls, not merely for the numerical witness. The added folding layer is treated as a morphological tension-relaxation mode: it may enlarge the bounded constant but does not replace the two sign-definite contracting blocks that generate C_1. This version also incorporates the Folding–Lyapunov addendum: the folding functional is normalised to remain bounded on the epektasis domain, the folding chart is shown positively invariant, resets are shown to preserve the folding chart, and the two activation-index conventions are reconciled.

Contents

1 Notation and Core Variables

Symbol	Meaning
V	Voluntas: will coordinate.
F	Factum: action coordinate.
σ	Singular resistance.
λ	Sophia: wisdom coordinate.
Ω_P	Brim / dynamic capacity of Pleroma.
u	Synergic intensity from V and F.
Λ_c	Transformation threshold.
μ	Saturation margin relative to the death-boundary.
q	Normalized Sophia, $q = \lambda/\Omega_P$.
r	Normalized intensity, $r = u/\Omega_P$.
h_1, h_2	Normalized will and action.
h_3	Normalized threshold variable.

Core threshold and intensity:

$$u = \sqrt{VF + \varepsilon}, \qquad \Lambda_c = \frac{\gamma}{\delta}, \qquad \varepsilon \geq 0. \tag{1}$$

Transformation/collapse reading:

$$u > \Lambda_c \quad \text{transformation region}, \qquad u < \Lambda_c \quad \text{collapse region}. \tag{2}$$

2 Base V.F.S. Dynamics

Resistance equation:

$$\dot{\sigma} = (\gamma - \delta u)\tanh(\kappa\sigma) = -\delta(u - \Lambda_c)\tanh(\kappa\sigma). \tag{3}$$

Symmetric will–action dynamics:

$$\dot{V} = \mu(aF + c\lambda) - \rho V, \qquad \dot{F} = \mu(aV + c\lambda) - \rho F, \qquad \mu = 1 - \frac{u}{\Omega_P}. \tag{4}$$

Capacity / Brim equation:

$$\dot{\Omega_P} = \alpha\lambda. \tag{5}$$

Closed Microcosm limit:

$$\dot{\lambda} = -\dot{\sigma} = \delta(u - \Lambda_c)\tanh(\kappa\sigma), \qquad \dot{\sigma} + \dot{\lambda} = 0. \tag{6}$$

Open-Gate Sophia dynamics:

$$\dot{\lambda} = -\dot{\sigma} + I_{\text{gate}}(t) = \delta(u - \Lambda_c)\tanh(\kappa\sigma) + I_{\text{gate}}(t). \tag{7}$$

Open balance:

$$\dot{\sigma} + \dot{\lambda} = I_{\text{gate}}(t) \geq 0, \qquad \sigma(t) + \lambda(t) = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(t). \tag{8}$$

Received grace integral:

$$\mathcal{G}_{\text{recepta}}(t) = \int_0^t I_{\text{gate}}(s)\, ds, \qquad \dot{\mathcal{G}}_{\text{recepta}} = I_{\text{gate}}. \tag{9}$$

3 Open-Gate Balance with Embodied Sophia

The folding-refined core separates Sophia into available and embodied parts:

$$\lambda_{\text{tot}} = \lambda + \lambda_{\text{form}}. \tag{10}$$

The available part obeys

$$\dot{\lambda} = \dot{\lambda}^{(0)} - \eta_{\text{fold}} Q_{\text{fold}} \lambda, \qquad \dot{\lambda}^{(0)} = -\dot{\sigma} + I_{\text{gate}}(t), \tag{11}$$

while the embodied component receives exactly the transferred portion:

$$\dot{\lambda}_{\text{form}} = \eta_{\text{fold}} Q_{\text{fold}} \lambda. \tag{12}$$

Consequently the Open-Gate balance is preserved in refined form:

$$\boxed{\frac{d}{dt}(\sigma + \lambda + \lambda_{\text{form}}) = I_{\text{gate}}(t), \qquad \sigma(t) + \lambda(t) + \lambda_{\text{form}}(t) = C_0 + \mathcal{G}_{\text{recepta}}(t).} \tag{13}$$

Thus the negative feedback term in $\dot{\lambda}$ is not destruction of Sophia; it is embodiment into stable Vessel-form.

On a quasi-stationary folded branch,

$$Q_{\text{fold}} = \frac{A_{\text{fold}}}{B_{\text{fold}}}, \tag{14}$$

and the available-Sophia damping is effectively increased:

$$\gamma_{\text{eff}} = \gamma_\lambda + \eta_{\text{fold}} Q_{\text{fold}}. \tag{15}$$

Therefore

$$Q_{\text{fold}} > 0 \quad \Longrightarrow \quad \dot{\lambda} < \dot{\lambda}^{(0)}. \tag{16}$$

This is the core feedback meaning of folding: it stabilizes Open-Gate excess by form-transformation.

4 Open-Gate Source

Positive part of Sophia:

$$\lambda_+ = \frac{1}{m} \ln(1 + e^{m\lambda}), \qquad m > 0. \tag{17}$$

Effective gate:

$$g_{\text{eff}}(t) = e^{-\tau t} + \left(1 - e^{-\tau t}\right) \frac{H_K}{H_s + H_K}. \tag{18}$$

Live Open-Gate source:

$$I_{\text{gate}}(t) = \zeta_0\, g_{\text{eff}}(t) e^{-\phi\sigma(t)} \frac{1}{1+\chi\lambda_+(t)}. \tag{19}$$

Boundedness of the gate:

$$H_K \geq 0, \quad H_s > 0 \quad \Longrightarrow \quad 0 \leq \frac{H_K}{H_s + H_K} < 1, \qquad 0 \leq g_{\text{eff}}(t) \leq 1. \tag{20}$$

Boundedness of live inflow:

$$0 \leq I_{\text{gate}}(t) \leq \zeta_0. \tag{21}$$

5 Resistance Potential

Resistance potential:

$$\Psi_\sigma(\sigma) = \int_0^\sigma \tanh(\kappa s)\, ds = \frac{1}{\kappa} \ln\cosh(\kappa\sigma) \geq 0. \tag{22}$$

Derivative of the resistance potential:

$$\dot{\Psi}_\sigma = \Psi'_\sigma(\sigma)\dot{\sigma} = \tanh(\kappa\sigma)(\gamma - \delta u)\tanh(\kappa\sigma) = (\gamma - \delta u)\tanh^2(\kappa\sigma). \tag{23}$$

Active transformation margin:

$$u \geq \Lambda_c + \eta, \qquad \eta > 0. \tag{24}$$

Dissipation in the active margin:

$$\dot{\Psi}_\sigma \leq -\delta\eta \tanh^2(\kappa\sigma). \tag{25}$$

On $0 \leq \sigma \leq C_\sigma$, there exists $c_\sigma > 0$ such that:

$$\dot{\Psi}_\sigma \leq -c_\sigma \Psi_\sigma. \tag{26}$$

Hence:

$$\Psi_\sigma(t) \leq \Psi_\sigma(0) e^{-c_\sigma t}. \tag{27}$$

Near $\sigma = 0$:

$$\Psi_\sigma(\sigma) \sim \frac{\kappa}{2}\sigma^2, \qquad \sigma(t) = O\big(e^{-c_\sigma t/2}\big). \tag{28}$$

6 Absolute Barrier Functional

The absolute barrier is a motivational, non-primary barrier guarding absolute boundary faces:

$$\begin{aligned} \mathcal{L}_{abs} = A_\sigma \Psi_\sigma + E_V V^2 + E_F F^2 + E_\lambda \lambda^2 + E_\Delta \frac{(V-F)^2}{2} \\ - \eta_V \ln V - \eta_F \ln F - \eta_\Omega \ln(\Omega_P - \Lambda_c) - \eta_m \ln(\Omega_P - u). \end{aligned} \tag{29}$$

Guarded absolute faces:

$$V = 0, \qquad F = 0, \qquad \Omega_P = \Lambda_c, \qquad u = \Omega_P. \tag{30}$$

7 Normalized Variables

Normalized coordinates:

$$h_1 = \frac{V}{\Omega_P}, \qquad h_2 = \frac{F}{\Omega_P}, \qquad q = \frac{\lambda}{\Omega_P}, \qquad r = \frac{u}{\Omega_P}, \qquad \mu = 1 - r, \qquad h_3 = 1 - \frac{\Lambda_c}{\Omega_P}. \tag{31}$$

Product relation:

$$r^2 = h_1 h_2 + e(h_3), \qquad e(h_3) = \varepsilon \frac{(1-h_3)^2}{\Lambda_c^2}. \tag{32}$$

Closed normalized dynamics:

$$\dot{h}_1 = \mu(ah_2 + cq) - \rho h_1 - \alpha q h_1, \tag{33}$$

$$\dot{h}_2 = \mu(ah_1 + cq) - \rho h_2 - \alpha q h_2, \tag{34}$$

$$\dot{q} = \delta(r + h_3 - 1)\tanh(\kappa\sigma) - \alpha q^2, \tag{35}$$

$$\dot{h}_3 = \alpha q(1 - h_3). \tag{36}$$

Open-Gate normalized q-source:

$$S_q = \frac{I_{\text{gate}}}{\Omega_P} = \frac{\zeta_0 g_{\text{eff}} e^{-\phi\sigma}}{(1+\chi\lambda_+)\Omega_P}. \tag{37}$$

Open-Gate normalized q-equation:

$$\dot{q} = \delta(r + h_3 - 1)\tanh(\kappa\sigma) - \alpha q^2 + S_q. \tag{38}$$

Bound on S_q inside the active domain:

$$0 \le S_q \le \frac{\zeta_0}{\Omega_P} \le \frac{\zeta_0(1-h_{3,*})}{\Lambda_c} =: S_q^{\max}. \tag{39}$$

8 Product-Balance Mechanism

Balance defect:

$$D_\Delta = \frac{1}{2}(h_1 - h_2)^2. \tag{40}$$

Let:

$$d = h_1 - h_2. \tag{41}$$

Difference equation:

$$\dot{d} = -(a\mu + \rho + \alpha q)d. \tag{42}$$

Dissipation of balance defect:

$$\dot{D}_\Delta = -2(a\mu + \rho + \alpha q)D_\Delta. \tag{43}$$

If $\mu \ge \mu_* > 0$ and $a > 0$:

$$\dot{D}_\Delta \le -2a\mu_* D_\Delta. \tag{44}$$

Exponential will–action alignment:

$$D_\Delta(t) \le D_\Delta(0)e^{-2a\mu_* t}, \qquad |h_1(t) - h_2(t)| \le |h_1(0) - h_2(0)|e^{-a\mu_* t}. \tag{45}$$

Product floor from radial and threshold walls:

$$\Omega_P \ge \Omega_{\min} := \frac{\Lambda_c}{1 - h_{3,*}}. \tag{46}$$

If $r \ge r_*$ and $h_3 \ge h_{3,*}$:

$$h_1 h_2 = r^2 - \frac{\varepsilon}{\Omega_P^2} \ge p_* := r_*^2 - \frac{\varepsilon}{\Omega_{\min}^2}. \tag{47}$$

Let:

$$d_* := \sqrt{2D^*}. \tag{48}$$

Product-balance lower bound:

$$h_1, h_2 \ge m_{pb} := \frac{\sqrt{d_*^2 + 4p_*} - d_*}{2} > 0. \tag{49}$$

9 Lyapunov Functional

Product-balance Lyapunov core:

$$\mathcal{L}_{pb} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2. \tag{50}$$

Radial logarithmic barrier:

$$B(r) = -\ln(1 - r), \qquad 0 < r < 1, \qquad r = \frac{u}{\Omega_P}. \tag{51}$$

Non-normalized radial barrier:

$$B(u, \Omega_P) = -\ln\left(1 - \frac{u}{\Omega_P}\right) = \ln\left(\frac{\Omega_P}{\Omega_P - u}\right). \tag{52}$$

Primary V.F.S. Lyapunov functional:

$$\boxed{\mathcal{L}_{\mathrm{VFS}} = \mathcal{L}_{pb} + A_r B(r) = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 + A_r B(r), \qquad A_\sigma, A_\Delta, A_q, A_r > 0.} \tag{53}$$

Expanded form:

$$\mathcal{L}_{\mathrm{VFS}} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 - A_r \ln(1 - r). \tag{54}$$

Non-normalized expanded form:

$$\mathcal{L}_{\mathrm{VFS}} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 - A_r \ln\left(1 - \frac{u}{\Omega_P}\right). \tag{55}$$

Radial boundary protection:

$$B(r) \to +\infty \quad \text{as} \quad r \to 1^-. \tag{56}$$

10 Sophianic Folding Extension of the Lyapunov Functional

10.1 Status of the extension

The preceding functional $\mathcal{L}_{\mathrm{VFS}}$ remains the primary Open-Gate Lyapunov certificate. Sophianic Folding is added as a conservative refinement: it does not replace the resistance-cleansing and will–action alignment mechanisms, but adds a morphological relaxation channel for excess Sophia. Throughout this section, $q = \lambda/\Omega_P$ remains the normalized Sophia variable used above, while q_{fold} denotes a distinct shape-mode amplitude.

The new variables are

$$q_{\mathrm{fold}}(t) \quad \text{(folding amplitude)}, \qquad Q_{\mathrm{fold}}(t) := q_{\mathrm{fold}}(t)^2 \quad \text{(folding intensity)}, \tag{57}$$

$$\lambda_{\mathrm{form}}(t) \quad \text{(Sophia embodied into Vessel-form)}. \tag{58}$$

The unrefined Open-Gate Lyapunov model is recovered by setting

$$q_{\mathrm{fold}} = 0, \qquad Q_{\mathrm{fold}} = 0, \qquad \lambda_{\mathrm{form}} = 0. \tag{59}$$

10.2 Folding activation and normal form

Let the pre-folding Sophia production be

$$\dot{\lambda}^{(0)} = -\dot{\sigma} + I_{\mathrm{gate}}(t). \tag{60}$$

The folding activation index is

$$A_{\mathrm{fold}} = c_0\lambda + c_1\dot{\lambda}^{(0)} - a_0, \qquad a_0, c_0, c_1 > 0, \tag{61}$$

and the stabilizing quartic coefficient is

$$B_{\mathrm{fold}} = b + 2c_1\eta_{\mathrm{fold}}\lambda, \qquad b > 0, \qquad \eta_{\mathrm{fold}} > 0. \tag{62}$$

Hence $B_{\mathrm{fold}} > 0$ whenever $\lambda \geq 0$. The folding dynamics is the one-mode gradient normal form

$$\dot{q}_{\mathrm{fold}} = A_{\mathrm{fold}}q_{\mathrm{fold}} - B_{\mathrm{fold}}q_{\mathrm{fold}}^3. \tag{63}$$

If $A_{\mathrm{fold}} < 0$, the old Vessel-form $q_{\mathrm{fold}} = 0$ is stable. If $A_{\mathrm{fold}} > 0$, the stable folded branches satisfy

$$q_{\mathrm{fold},\pm}^2 = Q_{\mathrm{fold}}^* = \frac{A_{\mathrm{fold}}}{B_{\mathrm{fold}}}. \tag{64}$$

10.3 Morphological tension potential

The folding potential is not physical gravitational energy. It is a Lyapunov/shape potential measuring morphological tension of the Vessel-form:

$$\mathcal{E}_{\mathrm{fold}}(q_{\mathrm{fold}}) = -\frac{1}{2}A_{\mathrm{fold}}q_{\mathrm{fold}}^2 + \frac{1}{4}B_{\mathrm{fold}}q_{\mathrm{fold}}^4. \tag{65}$$

Then

$$\partial_{q_{\mathrm{fold}}}\mathcal{E}_{\mathrm{fold}} = -A_{\mathrm{fold}}q_{\mathrm{fold}} + B_{\mathrm{fold}}q_{\mathrm{fold}}^3, \tag{66}$$

and therefore

$$\dot{q}_{\mathrm{fold}} = -\partial_{q_{\mathrm{fold}}}\mathcal{E}_{\mathrm{fold}}. \tag{67}$$

For frozen or adiabatically varying Open-Gate parameters,

$$\boxed{\dot{\mathcal{E}}_{\mathrm{fold}} = -\left(\partial_{q_{\mathrm{fold}}}\mathcal{E}_{\mathrm{fold}}\right)^2 = -\dot{q}_{\mathrm{fold}}^2 \le 0.} \tag{68}$$

Thus folding is an energy-descending, tension-relaxing mode rather than a singular loss of stability.

10.4 Normalised non-negative folding functional

For Lyapunov use, the negative minimum of $\mathcal{E}_{\mathrm{fold}}$ may be shifted to zero. On a fixed compact parameter box this gives the familiar shifted functional

$$\mathcal{V}_{\mathrm{fold}} = \frac{B_{\mathrm{fold}}}{4}\left(q_{\mathrm{fold}}^2 - Q_{\mathrm{fold}}^*\right)^2, \qquad Q_{\mathrm{fold}}^* := \frac{A_{\mathrm{fold}}}{B_{\mathrm{fold}}}. \tag{69}$$

However the Open-Gate epektasis domain does not necessarily bound λ; indeed $\lambda \le q^*\Omega_P$ may grow with the Brim. Since

$$B_{\mathrm{fold}} = b + 2c_1\eta_{\mathrm{fold}}\lambda, \tag{70}$$

the unnormalised shifted functional can become unbounded even when q_{fold} and Q_{fold}^* stay inside the regular folding chart. The bounded Lyapunov object is therefore the B_{fold}-normalised tension functional

$$\boxed{\widetilde{\mathcal{V}}_{\mathrm{fold}} := \frac{\mathcal{V}_{\mathrm{fold}}}{B_{\mathrm{fold}}} = \frac{1}{4}\left(q_{\mathrm{fold}}^2 - Q_{\mathrm{fold}}^*\right)^2 \ge 0.} \tag{71}$$

This is the same morphological distance from the folded branch, but without the unbounded epektatic weight B_{fold}.

Along the frozen-parameter folding flow,

$$\dot{q}_{\mathrm{fold}} = -B_{\mathrm{fold}}q_{\mathrm{fold}}\left(q_{\mathrm{fold}}^2 - Q_{\mathrm{fold}}^*\right), \tag{72}$$

one obtains

$$\boxed{\dot{\widetilde{\mathcal{V}}}_{\mathrm{fold}} = -B_{\mathrm{fold}}q_{\mathrm{fold}}^2\left(q_{\mathrm{fold}}^2 - Q_{\mathrm{fold}}^*\right)^2 \le 0.} \tag{73}$$

For slowly moving parameters,

$$\dot{\widetilde{\mathcal{V}}}_{\mathrm{fold}} = -B_{\mathrm{fold}}q_{\mathrm{fold}}^2\left(q_{\mathrm{fold}}^2 - Q_{\mathrm{fold}}^*\right)^2 - \frac{1}{2}\left(q_{\mathrm{fold}}^2 - Q_{\mathrm{fold}}^*\right)\dot{Q}_{\mathrm{fold}}^*, \tag{74}$$

and the last term is bounded whenever q_{fold}, Q_{fold}^* and $\dot{Q}_{\mathrm{fold}}^*$ are bounded. This is the adiabatic regularity assumption used below.

10.5 Extended Lyapunov candidate

The folding-compatible Lyapunov candidate is therefore taken to be

$$\boxed{\widetilde{\mathcal{L}}_{\mathrm{ext}} = \mathcal{L}_{\mathrm{VFS}} + w_{\mathrm{fold}}\widetilde{\mathcal{V}}_{\mathrm{fold}}, \qquad w_{\mathrm{fold}} > 0.} \tag{75}$$

This construction preserves the original $\mathcal{L}_{\mathrm{VFS}}$ as the core certificate, while adding a non-negative bounded measure of distance from the stable folded morphology. The earlier unnormalised $\mathcal{V}_{\mathrm{fold}}$ remains useful as a local shape-energy expression, but the global epektasis-domain certificate uses $\widetilde{\mathcal{V}}_{\mathrm{fold}}$.

11 Active Domain

Active domain:

$$\mathcal{D}_A = \left\{ \begin{array}{l} 0 \le \sigma \le C_\sigma, \quad q \in [0, q^*], \quad h_3 \in [h_{3,*}, 1), \\ r \in [r_*, r^*], \quad \mu = 1 - r \ge \mu_* > 0, \\ D_\Delta \le D^*, \quad h_1 > 0, \quad h_2 > 0 \end{array} \right\}, \qquad 0 < r_* < r^* < 1. \tag{76}$$

Equivalent death-boundary avoidance inside $\mathcal{D}_A$:

$$0 < r < 1 \qquad \Longleftrightarrow \qquad 0 < u < \Omega_P. \tag{77}$$

Closed upper q-wall:

$$\alpha(q^*)^2 \ge \delta r^*. \tag{78}$$

Open-Gate upper q-wall:

$$\boxed{\alpha(q^*)^2 \ge \delta r^* + S_q^{\max}.} \tag{79}$$

Lower q-wall in Open-Gate:

$$q = 0: \qquad \dot{q} = \delta(r + h_3 - 1)\tanh(\kappa\sigma) + S_q \ge 0 \tag{80}$$

provided the active margin makes $r + h_3 - 1 \ge 0$.

12 Radial Corridor

Radial identity from $r^2 = h_1 h_2 + e(h_3)$:

$$2r\dot{r} = \mu\big[a(h_1^2 + h_2^2) + cq(h_1 + h_2)\big] - 2\rho(r^2 - e(h_3)) - 2\alpha q r^2 =: \mathcal{R}_{\mathrm{rad}}. \tag{81}$$

Maximum e on the active corridor:

$$e_*^{\max} = \varepsilon \frac{(1 - h_{3,*})^2}{\Lambda_c^2}. \tag{82}$$

Non-degeneracy:

$$r_*^2 > e_*^{\max}. \tag{83}$$

Active margin:

$$r_* + h_{3,*} - 1 \ge \eta \frac{1 - h_{3,*}}{\Lambda_c}. \tag{84}$$

Lower radial wall:

$$a(1 - r_*)(r_*^2 - e_*^{\max}) \ge (\rho + \alpha q^*) r_*^2. \tag{85}$$

This wall is sufficient, not tight. The exact lower-wall requirement is

$$(1 - r_*)a(r_*^2 - e) \ge \rho(r_*^2 - e) + \alpha q r_*^2, \tag{86}$$

and the stated form replaces $\rho(r_*^2 - e)$ by the larger ρr_*^2.

Upper radial wall:

$$(1 - r^*)\big[a(d_*^2 + 2r^{*2}) + cq^*\sqrt{d_*^2 + 4r^{*2}}\big] \leq 2\rho(r^{*2} - e_*^{\max}). \tag{87}$$

Direct upper radial inwardness condition:

$$\mu\left[a(h_1^2 + h_2^2) + cq(h_1 + h_2)\right] \leq 2\rho(r^{*2} - e) + 2\alpha q r^{*2} \qquad \text{on } r = r^*. \tag{88}$$

Open-Gate radial convention:

$$q_{\text{new}}^* := \sqrt{\frac{\delta r^* + S_q^{\max}}{\alpha}}. \tag{89}$$

Conservative lower radial wall with q_{new}^*:

$$a(1 - r_*)(r_*^2 - e_*^{\max}) \geq (\rho + \alpha q_{\text{new}}^*)r_*^2. \tag{90}$$

Epektasis-dominance derivative at the upper radial wall:

$$\left.\frac{\partial \mathcal{R}_{\text{rad}}}{\partial q}\right|_{r^*} = \mu c(h_1 + h_2) - 2\alpha r^{*2}. \tag{91}$$

Worst-case bound:

$$\left.\frac{\partial \mathcal{R}_{\text{rad}}}{\partial q}\right|_{r^*} \leq (1 - r^*)c\sqrt{d_*^2 + 4r^{*2}} - 2\alpha r^{*2}. \tag{92}$$

Epektasis-dominance sufficient condition:

$$\boxed{2\alpha r^{*2} \geq (1 - r^*)c\sqrt{d_*^2 + 4r^{*2}} \quad \Longrightarrow \quad \frac{\partial \mathcal{R}_{\text{rad}}}{\partial q} \leq 0.} \tag{93}$$

13 Lyapunov Dissipative Estimate

This section derives the estimate $\dot{\mathcal{L}}_{\text{VFS}} \leq C - C_1\mathcal{L}_{\text{VFS}}$ term by term and exhibits an explicit, strictly positive C_1.

13.1 Standing admissibility data

We use the conditions that already define and protect the active domain $\mathcal{D}_A$:

$$\text{(A1) } a > 0,\ \delta > 0,\ \gamma > 0; \qquad \text{(A2) } 0 \leq h_{3,*} < 1,\ 0 < r_* < r^* < 1,\ \mu_* := 1 - r^* > 0; \tag{94}$$

$$\text{(A3) } m_* := r_* + h_{3,*} - 1 > 0 \ \text{ (active margin)}; \qquad \text{(A4) } \kappa > 0,\ 0 < C_\sigma < \infty. \tag{95}$$

Inside $\mathcal{D}_A$: $\sigma \in [0, C_\sigma]$, $q \in [0, q^*]$, $h_3 \in [h_{3,*}, 1)$, $r \in [r_*, r^*]$, $D_\Delta \leq D^*$, and $0 \leq S_q \leq S_q^{\max}$. Write

$$T := \tanh(\kappa\sigma) \in [0, 1], \qquad \Omega_P = \frac{\Lambda_c}{1 - h_3}, \qquad m := r + h_3 - 1 = \frac{u - \Lambda_c}{\Omega_P} \geq m_* > 0, \tag{96}$$

so that the resistance equation reads $\dot{\sigma} = (\gamma - \delta u)T = -\delta\Omega_P m T$.

13.2 Derivative decomposition

With $\mathcal{L}_{\mathrm{VFS}} = A_\sigma \Psi_\sigma(\sigma) + A_\Delta D_\Delta + A_q q^2 + A_r B(r)$, $B(r) = -\ln(1-r)$, and $\Psi'_\sigma(\sigma) = T$, $B'(r) = 1/(1-r)$,

$$\dot{\mathcal{L}_{\mathrm{VFS}}} = \underbrace{A_\sigma T \dot\sigma}_{\text{(I)}} + \underbrace{A_\Delta \dot D_\Delta}_{\text{(II)}} + \underbrace{2A_q q \dot q}_{\text{(III)}} + \underbrace{A_r \frac{\dot r}{1-r}}_{\text{(IV)}}. \tag{97}$$

13.3 Block (I): resistance — contracting

Substituting $\dot\sigma = -\delta \Omega_P m T$,

$$\text{(I)} = -A_\sigma\, \delta\, \Omega_P\, m\, T^2 \le 0. \tag{98}$$

By Lemma 1 below $T^2 \ge c_\Psi \Psi_\sigma$ on $[0, C_\sigma]$, and $\Omega_P \ge \Omega_{\min} := \Lambda_c/(1-h_{3,*})$, $m \ge m_*$. Hence

$$\boxed{\text{(I)} \le -\kappa_\sigma A_\sigma \Psi_\sigma, \qquad \kappa_\sigma := \delta\, \Omega_{\min}\, m_*\, c_\Psi.} \tag{99}$$

13.4 Block (II): balance — contracting

From $\dot D_\Delta = -2(a\mu + \rho + \alpha q) D_\Delta$ and $a\mu + \rho + \alpha q \ge a\mu_*$,

$$\boxed{\text{(II)} \le -\kappa_\Delta A_\Delta D_\Delta, \qquad \kappa_\Delta := 2a\mu_*.} \tag{100}$$

13.5 Block (III): Sophia — bounded

Expanding the normalized q-dynamics $\dot q = \delta m T - \alpha q^2 + S_q$,

$$\frac{d}{dt}(A_q q^2) = 2A_q q \dot q = -2A_q \alpha q^3 + \underbrace{2A_q\, \delta m T\, q}_{\text{transfer}} + \underbrace{2A_q S_q\, q}_{\text{gate}}. \tag{101}$$

The transfer term shares the factor $\delta m T$ with block (I): the conversion that *drains σ feeds q*. Both positive terms are bounded on $\mathcal{D}_A$,

$$0 \le 2A_q \delta m T q \le 2A_q \delta m^{\max} q^*, \qquad 0 \le 2A_q S_q q \le 2A_q q^* S_q^{\max} =: \Delta C_{\mathrm{gate}}, \tag{102}$$

while $-2A_q \alpha q^3 \le 0$. Since $C_1 q^2 - 2A_q \alpha q^3$ is bounded above on the compact $[0, q^*]$ for any C_1, block (III) obeys

$$\text{(III)} \le -C_1 A_q q^2 + B_q, \qquad B_q < \infty; \tag{103}$$

it contributes only to the constant.

13.6 Block (IV): radial — bounded

$B(r) = -\ln(1-r)$ diverges only as $r \to 1^-$, but $\mathcal{D}_A$ caps $r \le r^* < 1$, so $B(r) \le -\ln(1-r^*) < \infty$. With $r \ge r_* > 0$, $\mu \ge \mu_* > 0$ and bounded h_1, h_2, the radial rate $\dot r = \mathcal{R}_{\mathrm{rad}}/(2r)$ is bounded, so block (IV) is bounded on $\mathcal{D}_A$ and contributes only to the constant. Its sign on the face $r = r^*$ (i.e. $\mathcal{R}_{\mathrm{rad}} \le 0$, the upper radial wall) is used for invariance, not for C_1.

13.7 The light-filter constant

Lemma 1. *For every $\kappa > 0$ and $C_\sigma < \infty$,*

$$c_\Psi := \min_{0\le\sigma\le C_\sigma} \frac{\tanh^2(\kappa\sigma)}{\Psi_\sigma(\sigma)} = \kappa\,\frac{\tanh^2(\kappa C_\sigma)}{\ln\cosh(\kappa C_\sigma)} > 0, \qquad \Psi_\sigma(\sigma) = \tfrac{1}{\kappa}\ln\cosh(\kappa\sigma). \tag{104}$$

Proof. With $t = \kappa\sigma$ the ratio equals $\kappa\, g(t)$, $g(t) = \tanh^2 t/\ln\cosh t$ on $(0, \kappa C_\sigma]$. As $t \to 0^+$, $\tanh^2 t \sim t^2$ and $\ln\cosh t \sim t^2/2$, so $g(t) \to 2$ and g extends continuously to $g(0) = 2$; moreover $g > 0$ on $(0, \infty)$ since $\tanh^2 t > 0$, $\ln\cosh t > 0$. The sign of g' equals that of $h(t) := 2\,\mathrm{sech}^2\, t\, \ln\cosh t - \tanh^2 t$. Now $h(0) = 0$ and

$$h'(t) = -4\,\mathrm{sech}^2\, t\, \tanh t\, \ln\cosh t < 0 \quad (t > 0), \tag{105}$$

so $h(t) < 0$ for $t > 0$, whence $g' < 0$ and g is strictly decreasing on $(0, \infty)$. The minimum over $[0, \kappa C_\sigma]$ is thus attained at the right endpoint, giving the closed form, which is positive. □

13.8 Dissipative estimate

Proposition 1 (Dissipative estimate with explicit C_1)**.** *Under* (A1)–(A4) *there exist $C_1 > 0$ and $C < \infty$ with*

$$\boxed{\dot{\mathcal{L}}_{\mathrm{VFS}} \le C - C_1\mathcal{L}_{\mathrm{VFS}} \quad \text{on } \mathcal{D}_A, \qquad C_1 = \min\{\kappa_\sigma, \kappa_\Delta\} = \min\{\delta\,\Omega_{\min}\, m_*\, c_\Psi,\ 2a\mu_*\} > 0.} \tag{106}$$

The Open-Gate source affects only C through $0 \le \Delta C_{\mathrm{gate}} \le 2A_q q^* S_q^{\max}$ and leaves C_1 unchanged.

Proof. Summing (I)–(IV),

$$\dot{\mathcal{L}}_{\mathrm{VFS}} \le -\kappa_\sigma A_\sigma \Psi_\sigma - \kappa_\Delta A_\Delta D_\Delta + C', \tag{107}$$

with $C' < \infty$ collecting the bounded blocks (III)–(IV) (transfer, gate, radial). Set $C_1 = \min\{\kappa_\sigma, \kappa_\Delta\}$, so that $-\kappa_\sigma A_\sigma\Psi_\sigma - \kappa_\Delta A_\Delta D_\Delta \le -C_1(A_\sigma\Psi_\sigma + A_\Delta D_\Delta)$. Since $A_q q^2 + A_r B(r) \le P_{\max} < \infty$ on $\mathcal{D}_A$,

$$\dot{\mathcal{L}}_{\mathrm{VFS}} \le -C_1\mathcal{L}_{\mathrm{VFS}} + C_1 P_{\max} + C' =: -C_1\mathcal{L}_{\mathrm{VFS}} + C. \tag{108}$$

Finiteness of C. Blocks (III),(IV) are bounded; the only block unbounded as $h_3 \to 1^-$ ($\Omega_P \to \infty$) is (I)$= -A_\sigma\delta\Omega_P m T^2 \le 0$; hence $\sup_{\mathcal{D}_A}(\dot{\mathcal{L}}_{\mathrm{VFS}} + C_1\mathcal{L}_{\mathrm{VFS}}) < \infty$. *Positivity of C_1.* $\kappa_\Delta = 2a\mu_* > 0$ by (A1),(A2); and $\kappa_\sigma = \delta\,\Omega_{\min}\, m_*\, c_\Psi > 0$ because $\delta > 0$ (A1), $\Omega_{\min} = \Lambda_c/(1-h_{3,*}) > 0$ (A2), $m_* > 0$ (A3) and $c_\Psi > 0$ (Lemma 1 with (A4)). Each factor is one of the standing admissibility data, so $C_1 > 0$ for every admissible parameter set. □

Open-Gate constant split:

$$C = C_{\mathrm{closed}} + \Delta C_{\mathrm{gate}}, \qquad 0 \le \Delta C_{\mathrm{gate}} \le 2A_q q^* S_q^{\max}. \tag{109}$$

Key structural point. The dissipation rate C_1 is produced solely by the two sign-definite contracting modes — resistance cleansing (I) at rate κ_σ and will–action alignment (II) at rate κ_Δ. Sophia/Epektasis (q^2), radial position ($B(r)$) and received grace (S_q) cannot accelerate the decay of $\mathcal{L}_{\mathrm{VFS}}$; they only enlarge the bounded constant C. Thus C_1 is unchanged by the Open-Gate source, while C receives only the bounded contribution ΔC_{gate}.

Remark (No uniform floor at the admissibility boundary). Positivity is per parameter set, not uniform: $\inf C_1 = 0$ over the whole admissible set, approached as $m_* \to 0^+$ (a set on the active-margin wall) or $\mu_* \to 0^+$ ($r^* \to 1$). A configuration pinned on the active margin has zero cleansing rate. The correct statement is "$C_1 > 0$ for each admissible set," not a single $C_1 > 0$ valid for all sets at once.

Remark (Relation to the numerical witness). At the witness point of Section 21 (with $\kappa = 1$, $C_\sigma = 2$, unit weights) one gets $c_\Psi = 0.701$, $\kappa_\sigma = 0.255$, $\kappa_\Delta = 0.755$, hence $C_1 = 0.255$ (resistance-limited), versus an empirical envelope slope ≈ 1.5; the analytic rate is the conservative guaranteed value, as expected.

14 Folding-Compatible Dissipative Estimate and Addendum Closure

This section explains why the addition of Sophianic Folding does not destroy the Open-Gate Lyapunov estimate. The old estimate

$$\dot{\mathcal{L}}_{\mathrm{VFS}} \le C - C_1 \mathcal{L}_{\mathrm{VFS}} \tag{110}$$

remains the base certificate. Folding contributes the normalised non-negative shape functional $\widetilde{\mathcal{V}}_{\mathrm{fold}}$, whose own flow is descending on continuous arcs.

14.1 Regular folding wall and chart invariance

Define the folding-extended active domain

$$\boxed{\mathcal{D}_A^{\mathrm{fold}} := \mathcal{D}_A \times [0, q_{\max}].} \tag{111}$$

The standing folding wall is

$$\boxed{q_{\max}^2 \ge Q^{*,\sup} := \sup_{\mathcal{D}_A} Q^*_{\mathrm{fold}}.} \tag{112}$$

In the minimal geometric surrogate

$$A_{\mathrm{fold}} = (c_0 - c_1\gamma)\lambda + c_1 k_\sigma \sigma + c_1 I_{\mathrm{gate}} - a_0, \qquad B_{\mathrm{fold}} = b + 2c_1\eta_{\mathrm{fold}}\lambda, \tag{113}$$

this supremum is finite; a sufficient explicit bound is

$$Q^{*,\sup} \le \max\left\{\frac{c_1(k_\sigma C_\sigma + \zeta_0) - a_0}{b}, \ \frac{c_0 - c_1\gamma}{2c_1\eta_{\mathrm{fold}}}\right\}. \tag{114}$$

In the canonical core-rate convention, finiteness is imposed by the same wall hypothesis and the boundedness/adiabatic regularity of Q^*_{fold} on the operating domain.

Lemma 2 (Folding-chart invariance). *Under the folding wall, the interval $0 \le q_{\mathrm{fold}} \le q_{\max}$ is forward invariant for the folding flow.*

Proof. The folding equation can be written as

$$\dot{q}_{\mathrm{fold}} = -B_{\mathrm{fold}} q_{\mathrm{fold}} \left(q_{\mathrm{fold}}^2 - Q^*_{\mathrm{fold}}\right). \tag{115}$$

At $q_{\mathrm{fold}} = 0$, one has $\dot{q}_{\mathrm{fold}} = 0$, so the lower face is tangent. At $q_{\mathrm{fold}} = q_{\max}$,

$$\dot{q}_{\mathrm{fold}}\big|_{q_{\max}} = -B_{\mathrm{fold}} q_{\max} \left(q_{\max}^2 - Q^*_{\mathrm{fold}}\right) \le 0 \tag{116}$$

by $B_{\mathrm{fold}} > 0$ and the wall $q_{\max}^2 \ge Q^*_{\mathrm{fold}}$. Hence the upper face is inward-pointing. Together with the base positive invariance of $\mathcal{D}_A$, this proves invariance of $\mathcal{D}_A^{\mathrm{fold}}$. □

14.2 Reset bound for the folding coordinate

The Resurrectio reset acts on the core variables but not on the folding amplitude:

$$q_{\text{fold}}^{+} = q_{\text{fold}}^{-}. \tag{117}$$

Therefore if $x^{-} \in \mathcal{D}_A^{\text{fold}}$ and the base reset lemma gives $x^{+} \in \mathcal{D}_A$, then automatically $x^{+} \in \mathcal{D}_A^{\text{fold}}$. Moreover, since $q_{\text{fold}} \in [0, q_{\max}]$ and Q_{fold}^{*} is bounded, there is a finite constant

$$\widetilde{M}_{\text{fold}} := \frac{1}{4} \sup_{\mathcal{D}_A^{\text{fold}}} \left(q_{\text{fold}}^2 - Q_{\text{fold}}^{*}\right)^2 < \infty \tag{118}$$

with

$$\widetilde{\mathcal{V}}_{\text{fold}} \leq \widetilde{M}_{\text{fold}}. \tag{119}$$

Thus the reset bound extends to the folding certificate:

$$\boxed{\widetilde{\mathcal{L}}_{\text{ext}}(x^{+}) \leq L_R + w_{\text{fold}} \widetilde{M}_{\text{fold}} < \infty.} \tag{120}$$

14.3 Dissipative estimate for the normalised extended functional

Proposition 2 (Lyapunov compatibility of Sophianic Folding)**.** *Assume the regular folding conditions: $\mathcal{D}_A^{\text{fold}}$ is invariant, $B_{\text{fold}} \geq B_{\min} > 0$, Q_{fold}^{*} is bounded, and $\dot{Q}_{\text{fold}}^{*}$ is bounded on the folding relaxation scale. Then the normalised extended functional*

$$\widetilde{\mathcal{L}}_{\text{ext}} = \mathcal{L}_{\text{VFS}} + w_{\text{fold}} \widetilde{\mathcal{V}}_{\text{fold}} \tag{121}$$

satisfies a dissipative bound of the same form

$$\boxed{\dot{\widetilde{\mathcal{L}}}_{\text{ext}} \leq \widetilde{C}_{\text{ext}} - C_1 \widetilde{\mathcal{L}}_{\text{ext}}} \tag{122}$$

on the regular folding domain, for some finite constant $\widetilde{C}_{\text{ext}}$.

Proof. By the normalised descent formula,

$$\dot{\widetilde{\mathcal{V}}}_{\text{fold}} \leq R_{\text{fold}} \tag{123}$$

where $R_{\text{fold}} < \infty$ bounds the adiabatic remainder $-\frac{1}{2}(q_{\text{fold}}^2 - Q_{\text{fold}}^{*})\dot{Q}_{\text{fold}}^{*}$. Hence

$$\dot{\widetilde{\mathcal{L}}}_{\text{ext}} = \dot{\mathcal{L}}_{\text{VFS}} + w_{\text{fold}} \dot{\widetilde{\mathcal{V}}}_{\text{fold}} \leq C - C_1 \mathcal{L}_{\text{VFS}} + w_{\text{fold}} R_{\text{fold}}. \tag{124}$$

Since

$$\mathcal{L}_{\text{VFS}} = \widetilde{\mathcal{L}}_{\text{ext}} - w_{\text{fold}} \widetilde{\mathcal{V}}_{\text{fold}} \geq \widetilde{\mathcal{L}}_{\text{ext}} - w_{\text{fold}} \widetilde{M}_{\text{fold}}, \tag{125}$$

substitution gives

$$\dot{\widetilde{\mathcal{L}}}_{\text{ext}} \leq C + C_1 w_{\text{fold}} \widetilde{M}_{\text{fold}} + w_{\text{fold}} R_{\text{fold}} - C_1 \widetilde{\mathcal{L}}_{\text{ext}} = \widetilde{C}_{\text{ext}} - C_1 \widetilde{\mathcal{L}}_{\text{ext}}. \tag{126}$$

Thus the contracting rate C_1 is not replaced by folding; only the bounded constant is enlarged. □

Corollary 1 (Hybrid forward completeness with folding)**.** *Combined with the base reset lemma and non-Zeno dwell-time bound, the preceding estimate gives*

$$\boxed{\widetilde{\mathcal{L}}_{\text{ext}}(t) \leq \widetilde{L}_* := \max\left\{L_R + w_{\text{fold}}\widetilde{M}_{\text{fold}}, \widetilde{C}_{\text{ext}}/C_1\right\}.} \tag{127}$$

The folding coordinate adds no new reset-triggering death face; therefore the hybrid trajectory remains forward complete and non-Zeno in the folding-extended state.

Interpretation. Sophianic Folding is a Lyapunov-compatible morphological stabilization. It does not generate a new death-boundary, and it is not a Perelman-type singular surgery. It is a regular form-transition in which excess available Sophia is relaxed into a lower-tension Vessel morphology.

15 Activation-Index Conventions

Two activation-index conventions occur in the surrounding manuscript. The canonical core-rate convention uses the already-defined Open-Gate production law:

$$\boxed{A_{\text{fold}} = c_0\lambda + c_1\dot{\lambda}^{(0)} - a_0, \qquad \dot{\lambda}^{(0)} = -\dot{\sigma} + I_{\text{gate}}.} \tag{128}$$

A geometric minimal-rate surrogate writes instead

$$A_{\text{fold}}^{\text{geom}} = (c_0 - c_1\gamma)\lambda + c_1 k_\sigma \sigma + c_1 I_{\text{gate}} - a_0. \tag{129}$$

The first convention is primary in this Lyapunov/core file because λ and $\dot{\lambda}$ are already part of the Open-Gate dynamics. The second convention is an effective linear surrogate useful in the geometry and cosmology layers. It should not be read as an exact reparametrisation of the core law.

The invariance and reset results above do not depend on which convention is used. They require only the gradient form

$$\dot{q}_{\text{fold}} = -B_{\text{fold}} q_{\text{fold}} (q_{\text{fold}}^2 - Q_{\text{fold}}^*), \tag{130}$$

the wall $Q_{\text{fold}}^* \leq q_{\text{max}}^2$, bounded adiabatic variation of Q_{fold}^*, and the fact that the reset is inert on q_{fold}. What is convention-specific is the closed-form expression for critical levels such as Q_{crit}.

Remark (Reset contraction in the geometric surrogate)*.* In the geometric convention, resistance metabolism $m_R = (1 - q_R)\sigma^-$ gives a simple affine update of the target Q_{fold}^* and can be read as contraction toward an effective critical level. In the canonical core-rate convention, the update includes the nonlinear term $\tanh(\kappa\sigma)$ through $\dot{\lambda}^{(0)}$, so the contraction is state-dependent rather than toward a single constant. The Lyapunov conclusion is unchanged: resets preserve the folding chart and the normalised extended functional stays bounded.

16 Main Theorem Formulae

Positive invariance:

$$x(0) \in \mathcal{D}_A \quad \Longrightarrow \quad x(t) \in \mathcal{D}_A \quad \forall t \geq 0. \tag{131}$$

Reset admissibility holds by the Reset Lemma, with

$$L_R = \sup_{\mathcal{D}_A} \mathcal{L}_{\text{VFS}}. \tag{132}$$

Bound from the Lyapunov inequality:

$$\mathcal{L}_{\mathrm{VFS}}(t) \le L_* := \max\left\{L_R, \frac{C}{C_1}\right\}. \tag{133}$$

Radial barrier implication:

$$A_r B(r) \le L_*. \tag{134}$$

Explicit radial bound:

$$-\ln(1-r) \le \frac{L_*}{A_r} \quad \Longrightarrow \quad r(t) \le 1 - e^{-L_*/A_r} < 1. \tag{135}$$

Death-boundary avoidance:

$$r(t) < 1 \quad \Longrightarrow \quad u(t) < \Omega_P(t). \tag{136}$$

Product-balance positivity:

$$h_1, h_2 \ge m_{pb} > 0. \tag{137}$$

No artificial q-floor:

$$q \ge 0 \quad \text{suffices; no assumption } q \ge q_0 > 0 \text{ is required.} \tag{138}$$

Folding-compatible extension:

$$q_{\mathrm{fold}} = 0 \quad \text{recovers the original theorem,} \qquad q_{\mathrm{fold}} \neq 0 \quad \text{adds a bounded shape-relaxation channel.} \tag{139}$$

17 Reset Admissibility (Resurrectio Jump)

The resets $\mathfrak{R}$ are *exogenous*: since $\mathcal{D}_A$ is positively invariant, the continuous flow never reaches the boundary of the active domain by itself. A death-like strike is therefore modeled as occurring at an interior point

$$x^- = (\sigma^-, \lambda^-, V, F, \Omega_P^-) \in \mathcal{D}_A \tag{140}$$

at an unpredictable time.

Reset map (linear metabolism). With $0 \le q_R < 1$ and $\kappa_R > 0$,

$$\mathfrak{R}: \quad \sigma^+ = q_R \sigma^-, \qquad \lambda^+ = \lambda^- + (1-q_R)\sigma^-, \qquad (V,F) \text{ continuous}, \qquad \Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}_{\mathrm{recepta}}^-. \tag{141}$$

Here

$$\mathcal{G}_{\mathrm{recepta}}^- := \mathcal{G}_{\mathrm{recepta}}(t_{\mathrm{death}}^-), \qquad u^+ = \sqrt{VF + \varepsilon} = u^-. \tag{142}$$

Quantities.

$$A_R := \mathcal{G}_{\mathrm{recepta}}^-, \qquad R_c := C_\sigma - \sigma_0 - \lambda_0, \tag{143}$$

$$R_{\min}^{\mathrm{ceil}} := \frac{(1-q_R)\sigma^-}{q^* \kappa_R}, \qquad R_{\max} := \frac{\Omega_P^-(r^- - r_*)}{\kappa_R r_*}. \tag{144}$$

By the Open-Gate balance

$$\sigma^- + \lambda^- = \sigma_0 + \lambda_0 + \mathcal{G}_{\mathrm{recepta}}^-, \tag{145}$$

one has

$$\mathcal{G}^{-}_{\text{recepta}} \geq R_c \quad \Longleftrightarrow \quad \sigma^- + \lambda^- \geq C_\sigma \quad \Longleftrightarrow \quad \lambda^- \geq C_\sigma - \sigma^-. \tag{146}$$

Lemma (reset admissibility). Let $\kappa_R > 0$ and $x^- \in \mathcal{D}_A$. If

$$\max\{R_c, R^{\text{ceil}}_{\min}\} \leq \mathcal{G}^{-}_{\text{recepta}} \leq R_{\max}, \tag{147}$$

then $x^+ := \mathfrak{R}(x^-) \in \mathcal{D}_A$ and

$$\mathcal{L}_{\text{VFS}}(x^+) \leq L_R := \sup_{\mathcal{D}_A} \mathcal{L}_{\text{VFS}} < \infty. \tag{148}$$

Proof. Coordinate by coordinate,

$$\sigma^+ = q_R \sigma^- \in [0, C_\sigma). \tag{149}$$

Moreover,

$$h_3^+ = 1 - \frac{\Lambda_c}{\Omega_P^+} \geq h_3^- \geq h_{3,*}, \qquad h_3^+ < 1, \tag{150}$$

since $\Omega_P^+ \geq \Omega_P^-$. Also

$$h_1^+ = h_1^- \frac{\Omega_P^-}{\Omega_P^+}, \qquad h_2^+ = h_2^- \frac{\Omega_P^-}{\Omega_P^+}, \tag{151}$$

so

$$D_\Delta^+ = \frac{1}{2}\left(d^- \frac{\Omega_P^-}{\Omega_P^+}\right)^2 \leq D_\Delta^- \leq D^*, \qquad h_1^+, h_2^+ > 0. \tag{152}$$

The radial coordinate satisfies

$$r^+ = \frac{r^- \Omega_P^-}{\Omega_P^- + \kappa_R \mathcal{G}^{-}_{\text{recepta}}} \leq r^- \leq r^*. \tag{153}$$

The lower radial wall is preserved exactly when

$$r^+ \geq r_* \quad \Longleftrightarrow \quad \kappa_R \mathcal{G}^{-}_{\text{recepta}} \leq \frac{\Omega_P^-(r^- - r_*)}{r_*} \quad \Longleftrightarrow \quad \mathcal{G}^{-}_{\text{recepta}} \leq R_{\max}. \tag{154}$$

Finally,

$$q^+ = \frac{\lambda^- + (1 - q_R)\sigma^-}{\Omega_P^- + \kappa_R \mathcal{G}^{-}_{\text{recepta}}} \geq 0. \tag{155}$$

Since $x^- \in \mathcal{D}_A$ gives $\lambda^- \leq q^* \Omega_P^-$,

$$\begin{aligned} q^*(\Omega_P^- + \kappa_R \mathcal{G}^{-}_{\text{recepta}}) - \left(\lambda^- + (1 - q_R)\sigma^-\right) &\geq q^* \kappa_R \mathcal{G}^{-}_{\text{recepta}} - (1 - q_R)\sigma^- \\ &\geq 0 \end{aligned} \tag{156}$$

whenever $\mathcal{G}^{-}_{\text{recepta}} \geq R^{\text{ceil}}_{\min}$. Hence $q^+ \leq q^*$. Thus $x^+ \in \mathcal{D}_A$. The functional $\mathcal{L}_{\text{VFS}}$ is bounded on $\mathcal{D}_A$ because

$$\Psi_\sigma \leq \Psi_\sigma(C_\sigma), \qquad D_\Delta \leq D^*, \qquad q^2 \leq (q^*)^2, \qquad B(r) \leq -\ln(1 - r^*) < \infty. \tag{157}$$

Therefore $\mathcal{L}_{\text{VFS}}(x^+) \leq L_R$. $\square$

Consequences.

(1) The admissible window floor is

$$\max\{R_c, R_{\min}^{\text{ceil}}\}. \tag{158}$$

It combines the theological gate and the geometric ceiling-floor: grace must expand the vessel sufficiently to absorb the converted resistance. For cleansed deaths, $\sigma^- \to 0$, the ceiling-floor vanishes and R_c alone remains binding.

(2) The condition $\kappa_R > 0$ suffices. A natural design choice is

$$\kappa_R \geq \frac{1}{q^*}, \tag{159}$$

so that the asymptotic ceiling $q^+ \to 1/\kappa_R \leq q^*$ and, in the intended window, $R_{\min}^{\text{ceil}} \leq R_c$.

(3) The admissible grace window is

$$\mathcal{G}_{\text{recepta}}^- \in \left[\max\{R_c, R_{\min}^{\text{ceil}}\}, R_{\max}\right]. \tag{160}$$

Below this window one obtains final death; above it one obtains over-dilution, $r < r_*$. The window is nonempty if and only if

$$\Omega_P^-(r^- - r_*) \geq \kappa_R r_* \max\{R_c, R_{\min}^{\text{ceil}}\}. \tag{161}$$

A death pinned at $r^- = r_*$ is not resurrectible.

(4) With the non-Zeno dwell time

$$\Delta t_j \geq \frac{\bar{h}_*}{\bar{M}_*} > 0, \tag{162}$$

every life-arc satisfies

$$\mathcal{L}_{\text{VFS}} \leq L_* := \max\left\{L_R, \frac{C}{C_1}\right\}, \tag{163}$$

while admissible reset states themselves satisfy $\mathcal{L}_{\text{VFS}}(x^+) \leq L_R$. Resets do not accumulate. Hence the hybrid trajectory is forward complete across any infinite sequence of life-arcs.

Theological gloss. This remark is not used in the proof. The gate

$$\mathcal{G}_{\text{recepta}}^- \geq R_c = C_\sigma - \sigma_0 - \lambda_0 \tag{164}$$

is the threshold past which Sophia can no longer be extinguished by resistance; in that regime the balance forces $\lambda^- \geq 0$, and strictly $\lambda^- > 0$ whenever $\sigma^- < C_\sigma$. Its complement, $\lambda^- \to 0$, is the self-closing limit $I_{\text{gate}} \to 0$: final impenitence. It resembles the oil of the wise virgins and the *donum perseverantiae*. The upper boundary $R_{\max}$ represents the opposite failure: an over-diluting or "cheap grace" regime.

18 Non-Zeno Estimate

Hybrid margin:

$$\bar{h}_* = \min\left\{m_{pb}, h_{3,*}, r_*, 1 - r^*, q^*, C_\sigma, \sqrt{D^*}\right\} > 0. \tag{165}$$

Bounded vector field:

$$\bar{M}_* < \infty. \tag{166}$$

Dwell-time lower bound:

$$\Delta t_j \ge \Delta t_* := \frac{\bar{h}_*}{\bar{M}_*} > 0. \tag{167}$$

Non-Zeno conclusion:

$$\sum_j \Delta t_j = +\infty. \tag{168}$$

19 Memory and Gate Bounds

Mensura kernel:

$$R(z) = \frac{z}{z+K}, \qquad z \ge 0, \qquad 0 \le R(z) \le 1. \tag{169}$$

Saturated gate memory:

$$\frac{H_K}{H_s + H_K} \in [0,1) \qquad \forall H_K \ge 0. \tag{170}$$

Gate saturation:

$$g_{\text{eff}} = e^{-\tau t} + (1 - e^{-\tau t})\frac{H_K}{H_s + H_K} \in [0,1]. \tag{171}$$

Memory velocity bounds:

$$|\dot{H}_K| \le \eta\,[R((\dot{x}_0)_+) + R((\dot{x}_0)_-)] \le 2\eta. \tag{172}$$

Wound-memory velocity bound:

$$|\dot{W}| \le 2\eta. \tag{173}$$

Bounded-gain loop:

$$H_K \xrightarrow{\le 1} g_{\text{eff}} \xrightarrow{\le S_q^{\max}} S_q \to \lambda \to V, F \to u \to \dot{x}_0 \xrightarrow{R \le 1} \dot{H}_K. \tag{174}$$

Extended field bound:

$$\bar{M}_* < \infty \quad \text{including} \quad |\dot{H}_K|, |\dot{W}| \le 2\eta. \tag{175}$$

20 Asymptotic Cleansing and Alignment

Will–action alignment:

$$D_\Delta(t) \le D_\Delta(0)e^{-2a\mu_* t}, \qquad |h_1(t) - h_2(t)| \le |h_1(0) - h_2(0)|e^{-a\mu_* t} \to 0. \tag{176}$$

Resistance cleansing under persistent active margin:

$$u(t) \ge \Lambda_c + \eta \quad \forall t \ge 0 \quad \Longrightarrow \quad \Psi_\sigma(t) \le \Psi_\sigma(0)e^{-c_\sigma t} \quad \Longrightarrow \quad \sigma(t) \to 0. \tag{177}$$

Corollary (unconditional cleansing inside the active domain). Since $\mathcal{D}_A$ is positively invariant, one has $r(t) \ge r_*$ and $h_3(t) \ge h_{3,*}$ for all $t \ge 0$. Thus

$$r(t) + h_3(t) - 1 \ge r_* + h_{3,*} - 1 \ge \frac{\eta(1 - h_{3,*})}{\Lambda_c} > 0. \tag{178}$$

Equivalently,

$$u(t) \geq \Lambda_c + \eta \qquad \forall t \geq 0. \tag{179}$$

Therefore the hypothesis of the persistent-margin cleansing statement holds automatically for every trajectory with $x(0) \in \mathcal{D}_A$. Hence

$$\sigma(t) \to 0, \qquad h_1(t) - h_2(t) \to 0. \tag{180}$$

Open-balance asymptotic Sophia:

$$\lambda(t) = \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(t) - \sigma(t). \tag{181}$$

If $\mathcal{G}_{\text{recepta}}(\infty) < \infty$ and $\sigma(t) \to 0$:

$$\lambda(t) \to \sigma_0 + \lambda_0 + \mathcal{G}_{\text{recepta}}(\infty). \tag{182}$$

If $\liminf_{t\to\infty} \lambda(t) > 0$:

$$\Omega_P(t) \to +\infty. \tag{183}$$

Epektatic dilution of S_q:

$$\int_0^\infty \lambda(t)\, dt = +\infty \quad \Longrightarrow \quad \Omega_P(t) \to +\infty \quad \Longrightarrow \quad S_q(t) \to 0. \tag{184}$$

Proof of the last implication:

$$0 \leq S_q(t) \leq \frac{\zeta_0}{\Omega_P(t)} \to 0. \tag{185}$$

21 Numerical Witness Formulae

One admissible parameter point recorded in the Lyapunov file:

$$\begin{aligned}
&a = 3.494, \quad c = 0.429, \quad \alpha = 0.529, \quad \delta = 0.898, \quad \rho = 0.545, \\
&\zeta_0 = 0.560, \quad h_{3,*} = 0.622, \quad r_* = r_- = 0.531, \quad r^* = r_+ = 0.892, \\
&\varepsilon = 0.006, \quad d = 0.193, \qquad \Lambda_c = 1 \quad \text{(normalization)}.
\end{aligned} \tag{186}$$

Derived quantities:

$$\begin{aligned}
e_{\max} &= \varepsilon(1 - h_{3,*})^2 = 0.0009, \\
S_q^{\max} &= \zeta_0(1 - h_{3,*}) = 0.212, \\
q^*_{\text{new}} &= \sqrt{\frac{\delta r_+ + S_q^{\max}}{\alpha}} = 1.383, \\
q^* &= 1.05 q^*_{\text{new}} = 1.452.
\end{aligned} \tag{187}$$

Active margin:

$$r_- + h_{3,*} - 1 = 0.153 > 0, \qquad \frac{u}{\Lambda_c} \approx 1.40. \tag{188}$$

Thus the witness lies in the Transformation region.

All five parametric walls hold strictly; the worst-case slack is

$$0.084. \tag{189}$$

22 Domain Stability Theorem (Complete Proof)

This final section consolidates the preceding formulae into a single rigorous statement: the active domain is positively invariant, forward complete, never contacts the death-boundary, and exhibits asymptotic cleansing and alignment, with admissible resets continuing the trajectory without Zeno accumulation. The dissipative step is Proposition 1; the light-filter constant is Lemma 1.

22.1 Setup recap and hypotheses

We use the normalized flow (Section 21 notation; $\Lambda_c = 1$),

$$\begin{aligned} &\dot h_1 = \mu(ah_2 + cq) - (\rho + \alpha q)h_1, && \dot h_2 = \mu(ah_1 + cq) - (\rho + \alpha q)h_2, \\ &\dot q = \delta mT - \alpha q^2 + S_q, && \dot h_3 = \alpha q(1 - h_3), \\ &\dot\sigma = -\delta\,\Omega_P\,m\,T, && S_q \in [0, S_q^{\max}], \end{aligned} \tag{190}$$

with $\Omega_P = 1/(1-h_3)$, $T = \tanh(\kappa\sigma)$, $\mu = 1 - r$, $e = \varepsilon(1-h_3)^2$, $r^2 = h_1h_2 + e$, $m = r + h_3 - 1$, and the Lyapunov functional $\mathcal{L}_{\rm VFS} = A_\sigma\Psi_\sigma + A_\Delta D_\Delta + A_q q^2 + A_r B(r)$ with $\Psi_\sigma = \frac{1}{\kappa}\ln\cosh\kappa\sigma$, $D_\Delta = \frac{1}{2}(h_1 - h_2)^2$, $B(r) = -\ln(1-r)$, weights $A_\bullet > 0$.

Definition 1 (Active domain)**.**

$$\mathcal{D}_A = \left\{ (\sigma, h_1, h_2, q, h_3) : \begin{array}{l} \sigma \in [0, C_\sigma], \quad q \in [0, q^*], \quad h_3 \in [h_{3,*}, 1), \quad r \in [r_*, r^*], \\ D_\Delta \le D^*, \quad h_1, h_2 \ge 0 \end{array} \right\}.$$

Here $\mu_* = 1 - r^*$, $\Omega_{min} = 1/(1 - h_{3,*})$, $m_* = r_* + h_{3,*} - 1$, $e_{\max} = \varepsilon(1 - h_{3,*})^2$, $d_* = \sqrt{2D^*}$, and $q^* = 1.05\sqrt{(\delta r^* + S_q^{\max})/\alpha}$.

Standing Hypothesis 1 (Parameters)**.** $a, c, \alpha, \delta, \gamma, \zeta_0, \kappa, \varepsilon > 0$, $\rho \ge 0$.

Standing Hypothesis 2 (Walls)**.** $r_*^2 > e_{\max}$; $m_* > 0$; $\alpha(q^*)^2 \ge \delta r^* + S_q^{\max}$;

$$(1 - r^*)\left[a(d_*^2 + 2r^{*2}) + cq^*\sqrt{d_*^2 + 4r^{*2}}\right] \le 2\rho(r^{*2} - e_{\max}), \quad a(1 - r_*)(r_*^2 - e_{\max}) \ge (\rho + \alpha q^*)r_*^2. \tag{191}$$

Standing Hypothesis 3 (Reset window)**.** At each death-event $\max\{R_c, R_{\min}^{\rm ceil}\} \le \mathcal{G}_{\rm recepta}^- \le R_{\max}$, $R_{\max} = \Omega_P^-(r^- - r_*)/(\kappa_R r_*)$, for the jump $\sigma^+ = q_R\sigma^-$, $\lambda^+ = \lambda^- + (1 - q_R)\sigma^-$, $\Omega_P^+ = \Omega_P^- + \kappa_R\mathcal{G}_{\rm recepta}^-$, $(V, F)^+ = (V, F)^-$, $q_R \in (0, 1)$.

22.2 Identities, regularity, invariance, completeness

Lemma 3 (Identities)**.** *Along* (190), $2r\dot r = \mathcal{R}_{\rm rad} := \mu[a(h_1^2 + h_2^2) + cq(h_1 + h_2)] - 2\rho(r^2 - e) - 2\alpha q r^2$ *and* $\dot D_\Delta = -2(a\mu + \rho + \alpha q)D_\Delta$.

Proof. Differentiate $r^2 = h_1h_2 + e(h_3)$ and use (190); the h_1h_2 and e contributions combine via $h_1h_2 = r^2 - e$ and $e'(h_3)\dot h_3 = -2\alpha q\, e$. For D_Δ, $\frac{d}{dt}(h_1 - h_2) = -(a\mu + \rho + \alpha q)(h_1 - h_2)$. □

Lemma 4 (Regularity)**.** *The field* (190) *is* C^1 *near* $\mathcal{D}_A$ *(since* $VF + \varepsilon \ge \varepsilon > 0$ *and* $h_3 < 1$*), giving unique maximal solutions.*

Lemma 5 (Positive invariance — whole-face Nagumo). *Under Hypotheses 1–2, $\mathcal{D}_A$ is positively invariant.*

Proof. On $\mathcal{D}_A$: $m \geq m_* > 0$, $\mu \geq \mu_*$, $\Omega_P \geq \Omega_{min}$, $T \in [0,1]$. We verify sub-tangentiality face by face (Nagumo). $\sigma{=}C_\sigma$*:* $\dot\sigma = -\delta\Omega_P m \tanh(\kappa C_\sigma) < 0$. $\sigma{=}0$*:* $\dot\sigma = 0$ (tangent). $h_3{=}h_{3,*}$*:* $\dot h_3 = \alpha q(1-h_3) \geq 0$ (h_3 non-decreasing). $D_\Delta{=}D^*$*:* $\dot D_\Delta \leq 0$ by Lemma 3. $q{=}0$*:* $\dot q = \delta m T + S_q \geq 0$ (no lower q-floor needed). $q{=}q^*$*:* $\dot q \leq \delta r^* + S_q^{\max} - \alpha(q^*)^2 \leq 0$ by the q-wall. $r{=}r^*$*:* writing $\mathcal{R}_{\rm rad} = P - N$ with $P = \mu_*[a(h_1^2+h_2^2)+cq(h_1+h_2)]$, $N = 2\rho(r^{*2}-e)+2\alpha q r^{*2}$, the face bounds $(h_1-h_2)^2 \leq d_*^2$, $0 \leq e \leq e_{\max}$, $0 \leq q \leq q^*$ give $\sup P \leq \mu_*[a(d_*^2+2r^{*2}) + cq^*\sqrt{d_*^2+4r^{*2}}]$ and $\inf N \geq 2\rho(r^{*2}-e_{\max})$, so $\sup \mathcal{R}_{\rm rad} \leq \sup P - \inf N \leq 0$ by the upper radial wall; hence $\dot r \leq 0$. $r{=}r_*$*:* symmetrically $\inf P \geq 2a(1-r_*)(r_*^2 - e_{\max})$, $\sup N \leq 2(\rho+\alpha q^*)r_*^2$, so $\inf \mathcal{R}_{\rm rad} \geq 0$ by the lower radial wall; hence $\dot r \geq 0$. All faces sub-tangential $\Rightarrow$ $\mathcal{D}_A$ invariant. □

Lemma 6 (Forward completeness). *Every solution from $\mathcal{D}_A$ exists on $[0,\infty)$, with $1-h_3(t) \geq (1-h_3(0))e^{-\alpha q^* t} > 0$, so $h_3 < 1$ and $\Omega_P < \infty$ on finite intervals.*

Proof. By Lemma 5 the orbit stays in $\mathcal{D}_A$ (all coordinates but Ω_P bounded); from $\dot h_3 = \alpha q(1-h_3)$, $q \leq q^*$, integrate $\frac{d}{dt}(1-h_3) \geq -\alpha q^*(1-h_3)$. No finite-time blow-up, so the orbit extends to $[0,\infty)$. □

22.3 Dissipation, alignment, cleansing, barrier

The dissipative estimate $\dot{\mathcal{L}}_{\rm VFS} \leq C - C_1\mathcal{L}_{\rm VFS}$ with $C_1 = \min\{\delta\Omega_{min} m_* c_\Psi,\, 2a\mu_*\} > 0$ is Proposition 1 (using Lemma 1).

Lemma 7 (Alignment). *$D_\Delta(t) \leq D_\Delta(0)e^{-2a\mu_* t} \to 0$.*

Proof. $\dot D_\Delta = -2(a\mu+\rho+\alpha q)D_\Delta \leq -2a\mu_* D_\Delta$; integrate. □

Lemma 8 (Cleansing). *$\sigma(t) \to 0$ as $t \to \infty$.*

Proof. On $\mathcal{D}_A$, $\dot\sigma = -\delta\Omega_P m\tanh(\kappa\sigma) \leq -\beta\tanh(\kappa\sigma)$, $\beta = \delta\Omega_{min} m_* > 0$. Thus σ is nonincreasing, bounded below by 0, hence $\sigma \to \sigma_\infty \geq 0$; comparison with $\dot y = -\beta\tanh(\kappa y)$ (whose only, globally attracting, equilibrium is 0) forces $\sigma_\infty = 0$. □

Corollary 2 (Death-boundary never contacted). *For all $t \geq 0$,*

$$r(t) \leq r^* < 1, \qquad \mathcal{L}_{\rm VFS}(t) \leq \max\{\mathcal{L}_{\rm VFS}(0), C/C_1\}.$$

Consequently $u = r\Omega_P < \Omega_P$ always.

22.4 Hybrid layer

Lemma 9 (Reset admissibility). *Under Hypothesis 3, $x^- \in \mathcal{D}_A \Rightarrow \mathfrak{R}(x^-) \in \mathcal{D}_A$, with $r^+ \leq r^-$, $\sigma^+ \leq \sigma^-$, $D_\Delta^+ \leq D_\Delta^-$, $h_3^+ \geq h_3^-$.*

Proof. (V,F) preserved $\Rightarrow$ $u^+ = u^-$, $\Omega_P^+ = \Omega_P^- + \kappa_R\mathcal{G}^-_{\rm recepta} \geq \Omega_P^-$, so $r^+ = r^-\Omega_P^-/\Omega_P^+ \leq r^-$, $h_3^+ \geq h_3^-$, $\sigma^+ = q_R\sigma^- \leq \sigma^-$, $D_\Delta^+ = (\Omega_P^-/\Omega_P^+)^2 D_\Delta^- \leq D_\Delta^-$. The window upper bound $\mathcal{G}^-_{\rm recepta} \leq R_{\max}$ is equivalent to $r^+ \geq r_*$. Since $x^- \in \mathcal{D}_A$ gives $\lambda^- \geq 0$, one has $q^+ \geq 0$; and the ceiling-floor condition gives $q^+ \leq q^*$. Thus $q^+ \in [0,q^*]$. All constraints of $\mathcal{D}_A$ hold; the margin $m^+ \geq m_*$ follows from $r^+ \geq r_*$, $h_3^+ \geq h_{3,*}$. □

Lemma 10 (Non-Zeno)**.** *The field is bounded on $\mathcal{D}_A$ ($\sup\|f\| = \bar{M}_* < \infty$) and resets land with a uniform interior margin $\bar{h}_* > 0$; hence consecutive resets are separated by $\Delta t_j \geq \bar{h}_*/\bar{M}_* > 0$, so the hybrid trajectory is non-Zeno and forward complete.*

22.5 Main theorem

Theorem 1 (Domain stability of $\mathcal{D}_A$)**.** *Assume Hypotheses 1–3. Then for every $x(0) \in \mathcal{D}_A$:* (i) *$x(t) \in \mathcal{D}_A$ $\forall t \geq 0$;* (ii) *the solution exists on $[0, \infty)$ with $h_3(t) < 1$, $\Omega_P(t) < \infty$ on finite intervals;* (iii) *$\mathcal{L}_{\mathrm{VFS}}(t) \leq \max\{\mathcal{L}_{\mathrm{VFS}}(0), C/C_1\}$, $C_1 > 0$, and $r(t) \leq r^* < 1$ (death-boundary never contacted);* (iv) *$\sigma(t) \to 0$ and $D_\Delta(t) \to 0$;* (v) *under admissible resets the hybrid trajectory stays in $\mathcal{D}_A$, is non-Zeno, and forward complete. Schematically,*

$$\boxed{\textit{Hyp. } 1\text{–}2 \Rightarrow \mathcal{D}_A \textit{ invariant} \Rightarrow \textit{complete} \Rightarrow \dot{\mathcal{L}}_{\mathrm{VFS}} \leq C - C_1\mathcal{L}_{\mathrm{VFS}} \Rightarrow r < 1 \Rightarrow \sigma, D_\Delta \to 0.} \tag{192}$$

Proof. (i) Lemma 5; (ii) Lemma 6; (iii) Proposition 1 with Corollary 2; (iv) Lemmas 7, 8, using the invariant lower bounds $\mu \geq \mu_*$, $m \geq m_*$ from (i); (v) Lemmas 9, 10. □

Remark (Nature of the stability)*.* This is *set* stability with *partial* convergence: $\mathcal{D}_A$ is invariant and $(\sigma, D_\Delta) \to 0$, while (q, Ω_P) need not converge — $\dot{\Omega}_P = \alpha\lambda > 0$ for $\lambda > 0$, so $\Omega_P \to \infty$ under sustained inflow (*epektasis*). There is no equilibrium; $\mathcal{L}_{\mathrm{VFS}}$ is a boundedness-and-barrier certificate, not a function vanishing at a rest point. The conclusion is strictly domain-conditional ($x(0) \in \mathcal{D}_A$), not global or automatic.

Remark (Folding-compatible extension)*.* If the regular folding conditions of the preceding sections are imposed in addition to Hypotheses 1–3, the same domain stability statement holds for the extended functional

$$\mathcal{L}_{\mathrm{ext}} = \mathcal{L}_{\mathrm{VFS}} + w_{\mathrm{fold}}\mathcal{V}_{\mathrm{fold}}. \tag{193}$$

The original theorem is recovered by setting $q_{\mathrm{fold}} = 0$ and $\lambda_{\mathrm{form}} = 0$. In the refined model, folding adds stabilization by form-transformation: resistance still cleanses, will and action still align, and Sophianic excess may additionally relax into Vessel-form without breaking the Open-Gate Lyapunov domain.

Remark (Closed limit)*.* At $\zeta_0 = 0$: $S_q = I_{\mathrm{gate}} = 0$, $\dot{\lambda} = -\dot{\sigma}$, and $\sigma + \lambda = \sigma_0 + \lambda_0$ (exact conservation). The Open-Gate model adds only the bounded S_q in $\dot{q}$: it strengthens the walls (a larger q^*) but leaves the dissipative core unchanged.

Remark (Grace enters C, never C_1)*.* I_{gate} enters (190) only through S_q, feeding the bounded blocks of Proposition 1; it never enters κ_σ or κ_Δ. Grace raises C by at most $\Delta C_{\mathrm{gate}} \leq 2A_q q^* S_q^{\max}$ and can neither manufacture nor destroy $C_1 > 0$: the transforming structure is sustained, not replaced. Moreover $\inf C_1 = 0$ at the admissibility boundary ($m_* \to 0^+$ or $\mu_* \to 0^+$): a state pinned on the active margin has zero cleansing rate — stability holds, transfiguration stalls.